

Point Cloud Registration

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1 Introduction

The primary objective of this project is to gain hands-on experience in recent computer graphics research by implementing and evaluating deep learning-based 3D point cloud registration algorithms. We are motivated by the inherent limitations of purely geometry-based registration methods, which often fail in environments lacking geometric saliency, such as symmetrical objects or flat walls. By exploring both pure geometric and cross-modal approaches, we aim to understand how integrating radiometric (RGB) information can break geometric ambiguities and enhance the overall robustness of the registration process.

To achieve this, the task requires processing two partially overlapping 3D input point clouds—a source point cloud $\mathcal{P}$ and a target point cloud $\mathcal{Q}$ —to compute and output an optimal rigid transformation matrix $T = \{R, t\}$, which consists of a 3D rotation matrix $R \in SO(3)$ and a 3D translation vector $t \in \mathbb{R}^3$ for spatial alignment. In addressing this problem, our project tackles several critical technical challenges: overcoming ambiguous geometry in environments with symmetric similarities or extended planar structures where traditional networks struggle, establishing reliable correspondences under low-overlap constraints (e.g., 10%–30% co-visible areas), and achieving effective cross-modal integration without destroying 3D geometric structures.

For our evaluation, we use GeoTransformer as the baseline algorithm, which represents a highly efficient, purely geometry-based method capable of learning transformation-invariant geometric structures for RANSAC-free registration. As our state-of-the-art reference, we implement CoFF (Cross-modal Feature Fusion), an approach that improves upon geometric methods by using a two-stage fusion process to integrate 3D point cloud geometry with 2D RGB images. The performance of these algorithms is systematically compared across three standard datasets: 3DMatch, featuring high-overlap indoor scenes; 3DLoMatch, a challenging subset with low-overlap ratios between 10% and 30%; and IndoorLRS, an indoor dataset characterized by rich color textures and extended planar regions.

2 Materials and Methods

2.1 Materials

Experiments are conducted on three benchmark datasets. 3DMatch is a widely used benchmark collected using a Microsoft Kinect RGB-D sensor, and all test point cloud pairs have overlap ratios above 30%. To evaluate robustness in more difficult settings, we also use 3DLoMatch, which targets low-overlap pairs with overlap ratios between 10% and 30%.

In addition, we evaluate on IndoorLRS, collected using an Asus Xtion Live RGB-D sensor. Its point clouds are generated by fusing 100 consecutive frames, which provides richer color texture information and is suitable for testing multimodal feature fusion.

2.2 Methods

2.2.1 GeoTransformer.

As the baseline reference method, GeoTransformer (Geometric Transformer) aims to extract accurate correspondences for fast and robust point cloud registration. Unlike traditional keypoint-based methods, such as D3Feat, GeoTransformer adopts a detector-free paradigm. D3Feat successfully enables the joint learning of dense detection and description of 3D local features via a 3D fully convolutional network; however, detecting and matching repeatable sparse keypoints becomes extremely difficult in low-overlap scenarios. To tackle this challenge, GeoTransformer bypasses keypoint detection and instead seeks correspondences over down-sampled superpoints, which are subsequently propagated to dense points.

At its core, it proposes the Geometric Transformer module to learn contextual features that capture the geometric structure of point clouds by encoding pair-wise distances and triplet-wise angles. This design makes the learned features invariant to rigid transformations and highly robust in low-overlap cases. Due to its surprisingly high matching accuracy, GeoTransformer eliminates the need for RANSAC in the estimation of the alignment transformation, leading to a massive acceleration in the registration process.

2.2.2 Relational Geometric Constraint Attention (RGCA).

To mitigate geometric ambiguity and the heavy computational overhead inherent in GeoTransformer, we propose the Relational Geometric Constraint Attention (RGCA). Unlike conventional mechanisms that rely on absolute positional encodings, RGCA explicitly integrates 3D relative geometric transformations directly into the attention kernels, thereby embedding directional physical priors into the feature aggregation logic.

This structural induction allows the network to effectively disambiguate features in symmetric or co-planar topologies by exploiting global geometric context. By synchronizing with the backbone’s latent dimensions through a multi-head architecture, RGCA selectively prioritizes structurally salient keypoints, which bolsters matching robustness while simultaneously alleviating memory bottlenecks in large-scale point cloud analysis.

2.2.3 CoFF.

CoFF is a cross-modal feature fusion method that jointly uses 3D geometry and 2D RGB images for registration. It adopts KPConv-FPN as the 3D backbone to extract both superpoint and dense point features.

The method then applies two-stage feature fusion. In stage one, a 2D fully convolutional network extracts pixel-wise image features and assigns them to 3D input points, enhancing point features with global image context. In stage two, patch-wise image features are extracted and fused with superpoint features, preserving local texture details and reducing ambiguous geometry during coarse matching.

Finally, fused features are fed into a coarse-to-fine matching pipeline to establish correspondences and estimate the final rigid transformation.

3 Results

3.1 Validation of Reproduced Results

As summarized in Table 1, our reproduced results on 3DMatch and 3DLoMatch benchmarks are highly consistent with the original paper. Due to the utilization of a subset (2/3) of the 3DMatch dataset for evaluation, the performance slightly deviates from the official records; nonetheless, the overall performance remains highly competitive, effectively validating the successful reproduction of the GeoTransformer framework.

The reproduced results are highly consistent with those reported in the original paper, validating the effectiveness of CoFF. Across evaluated benchmarks, CoFF demonstrates strong registration performance.

3.2 Efficacy of Spatially-Inductive Interaction

The performance metrics in Table 2 indicate that our Spatially-Inductive Interaction (SII) upgrade consistently surpasses the baseline GeoTransformer. The integration of relative spatial transformations allows the model to achieve superior Inlier Ratios (IR), most notably on the 3DLoMatch benchmark where IR rose from 43.5% to 44.3%. This improvement underscores the capacity of the SII module to disambiguate geometric features in low-overlap or highly symmetric environments. Moreover, the steady gains in Registration Recall across 3DMatch (90.6%) and IndoorLRS (97.0%) demonstrate that embedding directional physical priors into the attention mechanism effectively refines correspondence quality, leading to a more robust and generalizable registration framework.

The fundamental driver behind these improvements is the explicit resolution of geometric ambiguity. Standard 1D positional encodings, originally designed for sequence modeling, often fail to adequately capture the complex 3D topologies inherent in point clouds. To address this, the SII module directly embeds relative 3D geometric transformations into the interaction calculations of Queries and Key-Value pairs. By injecting these directional and relative positioning priors, the network leverages richer global geometric contexts to differentiate highly similar local features, particularly in textureless planar or symmetric regions.

Furthermore, we ensure a seamless feature-space integration by strictly aligning the latent dimensions of the SII layer with the GeoTransformer backbone. Coupled with a 4-head attention configuration and a dropout rate of 0.1, this physically-guided mechanism forces the feature aggregation process to focus exclusively on structurally significant keypoints. This targeted interaction not only bolsters representational capacity but also partially alleviates the memory bottlenecks typically associated with standard global self-attention during dense point cloud processing.

3.3 Comparison with SOTA

To rigorously evaluate the generalization and transferability of the proposed framework, we first extend our experimental scope to the **IndoorLRS** dataset. As a widely recognized and demanding benchmark, IndoorLRS encompasses a vast array of diverse indoor

environments, characterized by varying structural scales and object densities. Testing on this dataset serves as a critical assessment of a model’s cross-dataset robustness and its ability to handle domain shifts without scene-specific fine-tuning.

Building upon our validated baseline and the SII enhancement, we evaluate the comprehensive performance of the CoFF framework against state-of-the-art (SOTA) methods across all benchmarks. As illustrated in Table 3, our reproduced results for CoFF are highly consistent with those reported in the original paper, fully validating the effectiveness of the algorithm. Across all evaluated datasets, the CoFF model exhibits superior registration performance. Particularly on the widely adopted 3DMatch dataset and the highly challenging 3DLoMatch dataset, the reproduced model achieves an impressive Registration Recall (RR) of 95.8% and 82.2%, respectively.

Furthermore, on the IndoorLRS benchmark, our model maintains top-tier performance, demonstrating a remarkable zero-shot capability. These metrics compellingly demonstrate that, compared to traditional pure-geometry approaches, the synergistic integration of cross-modal features and robust geometric modeling significantly elevates both the accuracy and success rate of point cloud registration. This proves that the structural patterns learned by our framework remain highly effective and generalizable even when transitioned to entirely novel and unseen environments.

4 Discussion

The experimental results highlight that while purely geometric methods such as GeoTransformer achieve strong performance under standard conditions, their reliance on geometric structure alone fundamentally limits their robustness in challenging scenarios. In particular, when the overlapping regions are dominated by symmetric structures or large planar surfaces, the discriminative power of geometric features is significantly weakened. Although the detector-free design and transformer-based contextual encoding improve correspondence matching, the model still struggles to resolve ambiguity when local geometric patterns lack uniqueness. Furthermore, the global attention mechanism, while effective, introduces non-negligible computational and memory overhead, especially for large-scale point clouds.

The proposed Spatially-Inductive Interaction (SII) module partially alleviates these issues by explicitly incorporating relative geometric transformations into the attention mechanism. By embedding directional and relational priors, the model gains improved capability in distinguishing structurally similar regions, as evidenced by consistent gains in inlier ratio and registration recall. However, this improvement remains incremental rather than fundamental. The method still operates within the paradigm of geometry-only reasoning, meaning that in scenarios where geometry itself is insufficiently informative, such as textureless indoor environments, the performance ceiling remains constrained. Additionally, introducing more complex attention formulations may further increase architectural complexity and sensitivity to hyperparameter tuning.

In contrast, CoFF demonstrates that cross-modal feature fusion provides a more effective solution to geometric ambiguity by leveraging complementary RGB information. The integration of pixel-wise and patch-wise image features enriches the representation of 3D points, enabling more reliable correspondence estimation,

Table 1: Results on 3DMatch and 3DLoMatch Datasets.

Method	3DMatch			3DLoMatch		
	FMR (%)↑	IR (%)↑	RR (%)↑	FMR (%)↑	IR (%)↑	RR (%)↑
GeoTransformer	97.9	64.4	90.5	87.7	43.5	74.1
GeoTransformer-pre-train	98.2	70.9	92.5	87.1	43.5	74.2

Table 2: Results on 3DMatch and 3DLoMatch Datasets (Performance Gain of RGC)

Method	3DMatch			3DLoMatch		
	FMR (%)↑	IR (%)↑	RR (%)↑	FMR (%)↑	IR (%)↑	RR (%)↑
GeoTransformer	97.9	64.4	90.5	87.7	43.5	74.1
GeoTransformer-RGC	97.6	65.3	90.6	88.3	44.3	74.2

Table 3: Results on 3DMatch, 3DLoMatch, and IndoorLRS Datasets (Comparison of CoFF)

Method	3DMatch			3DLoMatch			IndoorLRS		
	FMR (%)↑	IR (%)↑	RR (%)↑	FMR (%)↑	IR (%)↑	RR (%)↑	FMR (%)↑	IR (%)↑	RR (%)↑
GeoTransformer	97.9	64.4	90.5	87.7	43.5	74.1	97.9	76.4	96.4
CoFF	99.4	74.2	95.8	93.7	47.3	82.2	99.7	72.3	96.5

particularly in low-overlap and visually textured environments. Nevertheless, this advantage comes at the cost of reduced generality and increased system dependency. CoFF assumes accurate calibration and alignment between RGB images and point clouds, making it vulnerable to preprocessing errors. Moreover, its reliance on visual texture limits applicability in scenarios where RGB data is unavailable, unreliable, or lacks discriminative content.

Our experiments suggest that both geometric and cross-modal methods can break down when ambiguity exists simultaneously in both geometry and appearance, such as uniformly colored walls with repetitive structures. This indicates that neither geometric constraints nor low-level visual features are sufficient to fully resolve correspondence uncertainty in such conditions.

Future work may address these limitations by incorporating higher-level semantic understanding and additional sensing modalities. For example, integrating semantic features derived from pre-trained vision-language models could provide object-level discrimination beyond low-level texture and geometry. Similarly, incorporating temporal or motion cues, such as odometry or multi-frame consistency, may introduce additional constraints that stabilize registration in ambiguous regions. Another promising direction lies in designing lightweight yet expressive attention mechanisms that balance computational efficiency with representational power, enabling scalable deployment in real-world large-scale environments.